was decided that Eagle is much better for the design. Though it isn't an open source tool, it is free for non-commercial/academic applications and pretty much multi-platform.

Next step? A control software that is left on the ground—RocketView runs on a laptop, and is operated by the launching staff. This telemetry display software was originally written with GTK+ for visualising those events that come from the rocket via the radio link. Later, it was rewritten as a Java application with the same functionality, plus launch control functions. You can now safely push a button and a rocket will fly into the sky.

Figure 8: The PSAS community with its LV2 rocket

Fuel and further development

So what is the fuel that boosts the rocket engine? According to Sarah Sharp: "It's ammonium perchlorate." Greenberg elaborates, "We did indeed have a GOX/Paraffin engine project, but it's been on hold for several years now. We hope to restart that project sometime in the future. For now, we're sticking with very standard solid amateur rocketry motors: an ammonium perchlorate oxidizer with HTPB fuel. We launched on a small "N" motor this last May, and we're aiming for a larger "N" this October at the Black Rock Desert amateur rocketry launch. [13]"

Figure 7: The LV2 rocket on the tower, moments before launch

However, there is great potential in GOX/Paraffin and LOX/Paraffin technology. While it is absolutely safe to be used in rocketry, it is also much cheaper compared to any alternative. Besides, there is big research work on, which will allow hybrid motors with Active Fin Control (AFC), Thrust Vector Control (TVC), as well as a reaction control system (RCS) to use it. It could enable achieving a new flight record that's above the already-achieved altitude of 10 kilometres.

Greenberg reveals, "PSAS is going to reach higher altitudes sometime later next year. We're currently busy with rebuilding the avionics system, so you can subscribe to a mailing-list and help us reach the sky." END

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Resources

[1] http://en.wikipedia.org/wiki/Sputnik_1
[2] http://en.wikipedia.org/wiki/Sergey_Korolyov
[3] http://en.wikipedia.org/wiki/Wernher_von_Braun
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[5] http://en.wikipedia.org/wiki/V-2
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